

The One-Electron Universe Architecture: Emulating Unique Electron Properties

Version 2.0 — Revised & Scientifically Deepened | June 2026

Classification: Tardigradia Game Engine | Physics Deep Dive Series | TGPU v2.0 **Particle Class:** Lepton (Fermion) | Charge: $-e$ | Spin: $1/2$ **Source:** Extends v1.0 Emulating Unique Electron Properties.pdf (V1.0)

Δ Change Log V1.0 $\rightarrow$ V2.0

Parameter	V1.0	V2.0	Source
Electron g-factor precision	$g = 2.00231930436$ (V1.0 7 decimal places)	$g = 2.002\,319\,304\,360\,92 \pm 0.000\,000\,000\,000\,36$ (CODATA 2022, 13 decimal places — most precise QED test ever)	CODATA 2022; Harvard g-2 (Hanneke et al., Nature 2023 validation)
Pines Demon quasiparticle	Not covered in V1.0	Confirmed experimentally 2023 in SrVO ₃ (strontium vanadate) by Kogar group (UIUC, Nature 2023). Massless, chargeless acoustic plasmon — second distinct electron oscillation mode discoverable in worldline simulation	Kogar et al., Nature 624 (2023)
Electron EDM constraint	EDM limit from V1.0 era:	de	$< 8.7 \times 10^{-29} \text{ e} \cdot \text{cm}$
Topological Berry phase implementation	Berry phase as static geometric phase value	V2.0: Time-varying Berry curvature (Berry tensor $\Omega_{\{ij\}}$) — electron's anomalous velocity $v_{\text{anomalous}} = (e/\hbar) \mathbf{E} \times \boldsymbol{\Omega}$ tracked per worldline fold. Relevant to topological insulator emulation	Xiao et al., Rev. Mod. Phys. 82 (2010); Armitage 2023 review
WebGPU compute shader	WebGL/GLSL vertex shaders for particle update	V2.0: WebGPU (WGSL) compute shaders for worldline integration. Chrome 113+ stable (May 2023). Enables 10^7 particle instances vs 10^3 in WebGL	WebGPU W3C Spec 2023; MDN Chrome 113 release

Parameter	V1.0	V2.0	Source
Spinor rotation in JS engine	4pi rotation noted conceptually	V2.0: Quaternion-based spinor SU(2) rotation: q_spin = [cos(theta/2), n_xsin(theta/2), n_ysin(theta/2), n_z*sin(theta/2)]. Implemented as Hamilton product in particle update loop	Dirac spinor formalism; Three.js Quaternion class

1. The One-Particle Universe Framework — V2.0 Update

The foundational architecture established in V1.0 — Worldline Monism, the “Origami Universe” metaphor, and the Object-Oriented ontology distinguishing intrinsic from extrinsic properties — is **fully preserved** and remains physically rigorous. V2.0 layers NEW discoveries from 2022-2026 atop this framework.

1.1 WebGPU Compute Architecture (V2.0 Core Infrastructure)

V1.0 designed for WebGL/Three.js on the CPU-side particle update loop. V2.0 upgrades to WebGPU (available in Chrome 113+ May 2023, Firefox 2024):

```
// WGSL Compute Shader - V2.0 Particle Integration
@binding(0) @group(0) var<storage, read_write> particles: array<ParticleState>;

struct ParticleState {
    position: vec4f,
    momentum: vec4f,
    spin: vec2f,           // Spinor components
    berry_phase: f32,     // Geometric phase accumulated
    proper_time: f32,     // tau - worldline parameter
    color_charge: u32,     // Packed color in 2 bits (R=0,G=1,B=2)
    quantum_numbers: u32, // Packed: isospin, strangeness, charm, beauty
    decay_lifetime: f32,
    entanglement_id: i32, // -1 = unentangled; >=0 = entanglement group
}

@compute @workgroup_size(64)
fn updateParticles(@builtin(global_invocation_id) id: vec3u) {
    let i = id.x;
    if (i >= arrayLength(&particles)) { return; }

    var p = particles[i];

    // Feynman-Schwinger proper time evolution
    let dtau = uniforms.dt / p.mass; // proper time step

    // Update worldline parameter
    p.proper_time += dtau;
```

```

// Berry phase accumulation (topological phase)
p.berry_phase += dot(uniforms.gauge_field, p.momentum.xyz) * dtau;

// Lorentz force (for charged particles)
let F = cross(uniforms.B_field, p.momentum.xyz) * p.charge_sign;
p.momentum.xyz += F * uniforms.dt;
p.position.xyz += p.momentum.xyz * uniforms.dt / p.energy;

particles[i] = p;
}

```

1.2 Updated Physical Constants (CODATA 2022)

Constant	V1.0 Value	V2.0 Value (CODATA 2022)
m _e	9.10938356×10 ⁻³¹ kg	9.1093837139×10 ⁻³¹ kg
e	1.60217663×10 ⁻¹⁹ C	1.602176634×10 ⁻¹⁹ C (exact)
ħ	1.054571817×10 ⁻³⁴ J·s	1.054571817×10 ⁻³⁴ J·s (exact)
c	2.99792458×10 ⁸ m/s	2.99792458×10 ⁸ m/s (exact)
α	1/137.036	1/137.035999084 ±0.000000021

2. V2.0 Enhanced Data Structures

Extending V1.0 struct definitions with 2022-2026 discoveries:

```

// V2.0 Universal Particle State – extends V1.0
struct ParticleInstanceV2 {
    // Inherited from V1.0
    double proper_time;           // tau worldline parameter (meter)
    glm::dvec3 position;          // 3-space vector
    glm::dvec3 momentum;          // 3-momentum (relativistic)
    float spin_z;                 // +0.5 or -0.5 (or eigenvalue)
    float berry_phase;            // Accumulated Berry/Aharonov-Bohm phase

    // V2.0 ADDITIONS
    float entanglement_r;         // Reduced density matrix purity 0-1
    int entanglement_partner_id;  // -1 = none; global particle ID
    float flavor_oscillation_phase; // For neutrinos/kaons
    uint32_t color_bits;          // 3-bit: R(b0),G(b1),B(b2)
    float decay_probability_dt;    // P(decay in dt) for unstable particles
    float vacuum_alignment;        // Order parameter for condensate mode

    // WebGPU Buffer packing
    // Total: ~100 bytes/particle * 10^6 particles = 100 MB GPU memory
    float _pad[2];                // Align to 16 bytes for WGSL
};

```

5. V2.0 — Pines’ Demon: The Second Electron Mode

The 2023 experimental confirmation of Pines’ Demon in SrVO₃ (strontium vanadate) demands a fundamental extension to the One-Electron Universe data structure. Pines predicted in 1956 that in multi-band metals, a second plasmon mode could exist where electrons from different bands oscillate 180° out of phase, cancelling their charge signals — creating a massless, electrically neutral acoustic plasmon.

Structural implication for OEU simulation:

The demon is not a separate particle — it is an interference pattern between two worldline branches. In OEU terms, when the single electron’s worldline intersects the “Now” slice twice at nearby spatial points, and those two intersection points (one from each energy band) are exactly pi radians out of phase, their electric fields cancel. The resulting observable is a “Demon Instance” — an entity invisible to photons but detectable via energy-loss spectroscopy.

```
struct DemonInstance {
    // Emerges when two ElectronInstances satisfy demon condition
    float phase_difference;           // Must be ~pi rad for demon formation
    float band_1_tau;                 // Proper time of first worldline intersection
    float band_2_tau;                 // Proper time of second intersection
    float acoustic_velocity;          //  $v_D \sim 0.4 * v_{Fermi}$  in SrVO3
    bool is_topologically_stable;     // Stability condition

    static bool isDemon(ElectronInstance& e1, ElectronInstance& e2) {
        float dPhase = std::abs(e1.berry_phase - e2.berry_phase);
        return (std::abs(dPhase - M_PI) < 0.05f) && (e1.band_index != e2.band_index);
    }
};
```

Rendering: Demon instances rendered as deep purple/void particles — present but invisible to the photon-based visual system. Revealed only when a special “spectroscopy probe” tool scans the worldline density.

6. V2.0 — ACME III EDM Constraint and CP Violation Topology

The ACME III result (2023) constrains the electron’s electric dipole moment to $|d_e| < 4.1 \times 10^{-30}$ e·cm. This means the electron’s charge distribution is symmetric to within 10^{-30} cm — implying the worldline has no intrinsic “twist” in the spatial direction that would create an EDM.

For OEU simulation: Zero EDM means the worldline knot at each intersection is topologically a “proper” knot with left-right symmetry. Any EDM-like term in the simulation represents BSM (Beyond Standard Model) physics — a “corruption” of the worldline topology for game mechanics.

```
struct EDMTopology {
    static constexpr double EDM_UPPER_BOUND = 4.1e-30; // e*cm (ACME III 2023)
    double d_e; // Electron EDM (zero in SM, nonzero in BSM scenarios)

    // "Corruption factor" for game mechanics
    float bsm_amplitude = 0.0f; // 0 = pure SM electron; 1.0 = maximally corrupted
};
```

8. V2.0 Benevolence Metric Update

The “Benevolence” concept central to all OEU documents is given a precise V2.0 mathematical definition grounded in 2022-2026 quantum information theory:

Benevolence = Quantum Coherence + Information Preservation + Symmetry Maintenance

$$B(t) = C(t) \cdot F(t) \cdot S(t)$$

Where: - $C(t)$ = l_1 -norm coherence of particle density matrix (Baumgratz et al. 2014) - $F(t)$ = quantum Fisher information / classical Fisher information (ratio ≥ 1) - $S(t)$ = fraction of symmetries unbroken at time t

V2.0 game mechanic: Benevolence score increases when: - Particles maintain long entanglement coherence (high C) - Topological protections are active (Berry phase winding conserved) - Symmetry groups are preserved in interactions (no spurious symmetry breaking)

*End of v1.0 Emulating Unique Electron Properties V2.0 | Tardigradia Game Engine SubParticles Series
This document supersedes V1.0 in all physics specifications.*